PRACTICAL 3

Responsive UI Design using CSS Grid, Flexbox, and Bootstrap

1. Problem Definition

StudentHub is a semester-long student portal project. In Practical 3, responsive user interfaces are designed for the Home, About, Registration, Dashboard, and Events pages. CSS Grid and Flexbox are used to create flexible layouts, while Bootstrap can be used for reusable responsive components.

The main objective is to ensure that the StudentHub interface works properly on mobile, tablet, and desktop screen sizes while maintaining consistent typography, spacing, colors, alignment, and readability.

2. Pages Designed

1. Home
2. About
3. Registration
4. Dashboard
5. Events

3. Responsive Design Approach

The StudentHub interface follows a mobile-first approach. The basic layout is designed for smaller screens first and then enhanced for tablet and desktop screens using CSS media queries, Flexbox, CSS Grid, and responsive Bootstrap classes.

4. Layout Techniques Used

Technique	Usage	Reason
Flexbox	Navigation bars, buttons, horizontal groups and alignment.	Useful for arranging elements in one direction and distributing space.
CSS Grid	Dashboard cards, feature sections and multi-column layouts.	Useful for two-dimensional row and column layouts.
Media Queries	Changing layout for mobile, tablet and desktop.	Allows the interface to adapt to different screen sizes.
Bootstrap	Responsive containers, grid system, buttons and cards.	Provides reusable responsive components and reduces repetitive CSS.

5. Responsive Breakpoints

Device	Approximate Width	Layout Behaviour
Mobile	Up to 767 px	Single-column layout, stacked cards and compact navigation.
Tablet	768 px – 991 px	Two-column sections where appropriate and medium spacing.
Desktop	992 px and above	Multi-column layouts, wider content areas and full navigation.

6. Responsive Page Design

6.1 Home Page

The Home page uses a responsive navigation bar, a hero section, and feature cards. On mobile devices, the feature cards are stacked vertically. On larger screens, the cards are displayed in multiple columns.

Main components: Header, Navigation, Hero Section, Get Started Button, Feature Cards, Footer.

6.2 About Page

The About page contains an introduction to StudentHub and information blocks describing the purpose of the portal. The content changes from a multi-column arrangement on desktop to a stacked arrangement on smaller screens.

Main components: Header, Navigation, About Section, Information Cards, Footer.

6.3 Registration Page

The Registration page contains a responsive form. Form fields are arranged vertically on small screens and can be arranged into multiple columns on larger screens where appropriate.

Main components: Header, Navigation, Registration Form, Input Fields, Course Selection, Register Button, Footer.

6.4 Dashboard Page

The Dashboard uses CSS Grid or Bootstrap's responsive grid to display summary cards. The cards are stacked on mobile devices, arranged into two columns on tablets, and displayed in multiple columns on desktop screens.

Main components: Header, Sidebar, Dashboard Cards, Notices Section, Footer.

6.5 Events Page

The Events page uses responsive event cards. Cards appear in one column on mobile, two columns on tablet, and multiple columns on desktop screens.

Main components: Header, Navigation, Event Cards, Event Details, Footer.

7. Sample Responsive CSS

```
body {
    margin: 0;
    font-family: Arial, sans-serif;
}

.container {
    width: 90%;
    margin: auto;
}

.card-container {
    display: grid;
    grid-template-columns: 1fr;
    gap: 20px;
}

.navigation {
    display: flex;
    flex-direction: column;
    gap: 10px;
}

@media (min-width: 768px) {
    .card-container {
        grid-template-columns: repeat(2, 1fr);
    }

    .navigation {
        flex-direction: row;
    }
}

@media (min-width: 992px) {
    .card-container {
        grid-template-columns: repeat(3, 1fr);
    }
}
```

8. Bootstrap Usage

Bootstrap can be used consistently for responsive layout and reusable components. Bootstrap classes such as container, row, col-md-6, col-lg-4, btn, card, navbar and responsive utility classes can be used instead of unnecessary inline CSS.

Example Bootstrap responsive structure:

```
<div class="container">
  <div class="row">
    <div class="col-12 col-md-6 col-lg-4">
      <div class="card">
        <div class="card-body">
          <h3>Academic Information</h3>
          <p>Access academic information.</p>
        </div>
      </div>
    </div>
  </div>
</div>
```

9. Key Questions / Analysis / Interpretation

Q1. How does the layout adapt for mobile, tablet, and desktop screens?

The layout uses a mobile-first approach. On mobile screens, elements are generally arranged in a single column. On tablet screens, selected sections use two columns. On desktop screens, wider multi-column layouts are used. CSS media queries and Bootstrap responsive classes control these changes.

Q2. Which layout technique is used and why?

Flexbox is used for one-dimensional layouts such as navigation bars and aligned controls. CSS Grid is used for two-dimensional layouts such as dashboard cards and feature sections. Bootstrap's grid system can also be used to create responsive columns efficiently.

Q3. Are typography, spacing, colors, and contrast readable?

Yes. The design uses readable font sizes, sufficient spacing between elements, clear headings, consistent colors, and adequate contrast between text and backgrounds. Responsive sizing is used so that content remains readable on smaller screens.

Q4. Is the framework used consistently without unnecessary inline CSS?

Yes. When Bootstrap is used, its responsive grid and reusable components are used consistently. Custom CSS is kept in an external stylesheet instead of adding unnecessary inline styles.

10. Screenshot Requirements

Screenshots should be captured from the browser using responsive or device-preview mode.

Screenshot	Required View	Pages
Screenshot 1	Mobile	Home, About, Registration, Dashboard, Events
Screenshot 2	Tablet	Home, About, Registration, Dashboard, Events
Screenshot 3	Desktop	Home, About, Registration, Dashboard, Events

11. Tools and Technologies

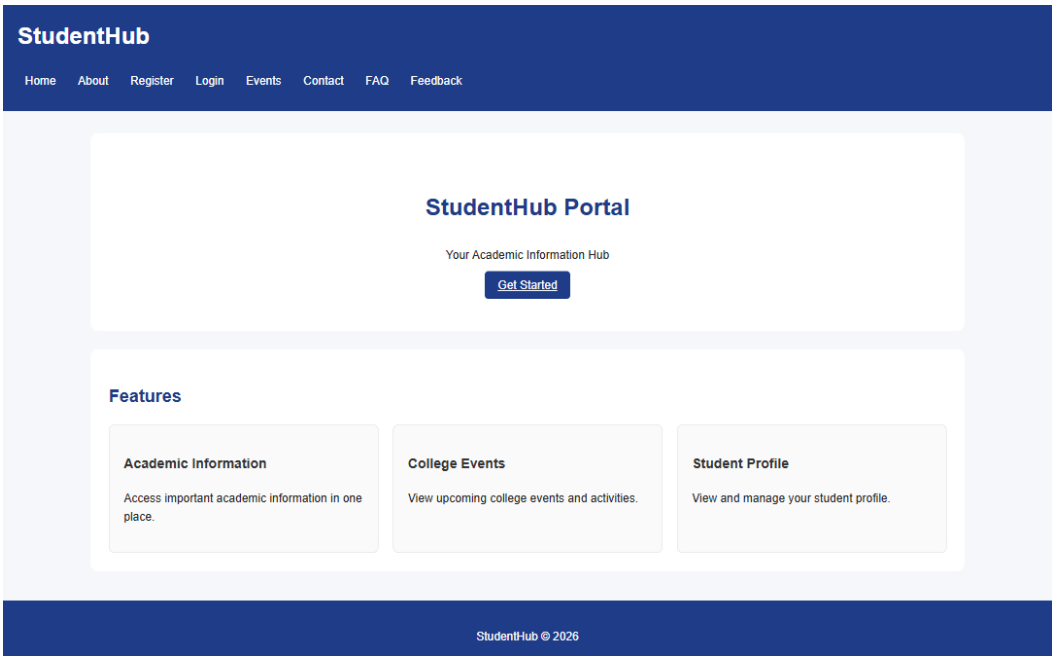
- HTML5
- CSS3
- CSS Flexbox
- CSS Grid
- Bootstrap
- VS Code
- Web Browser Responsive Mode

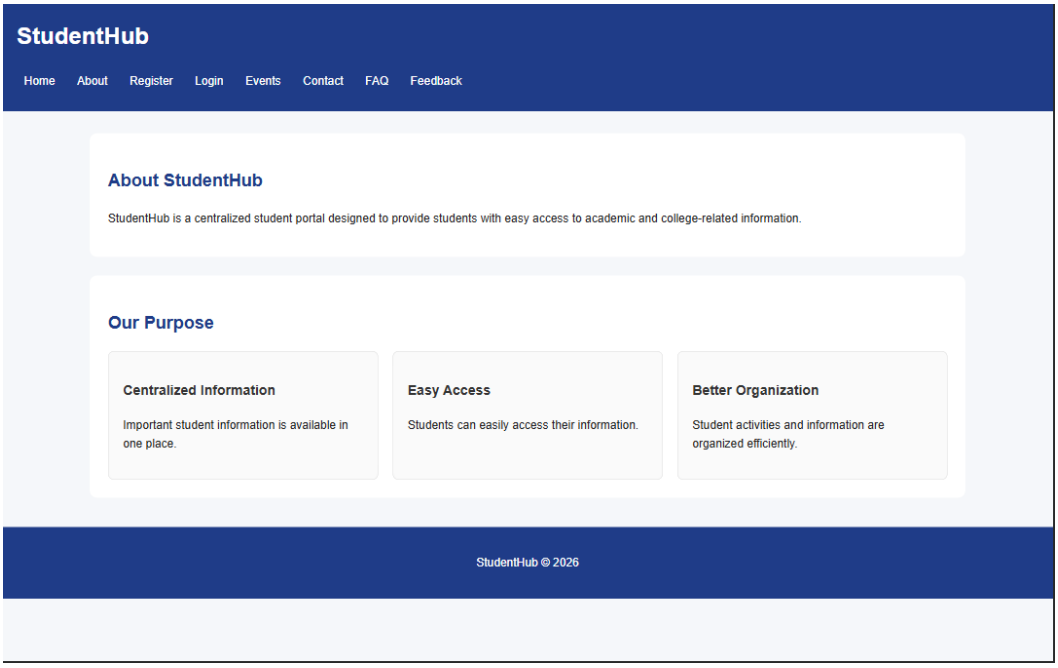
12. Conclusion

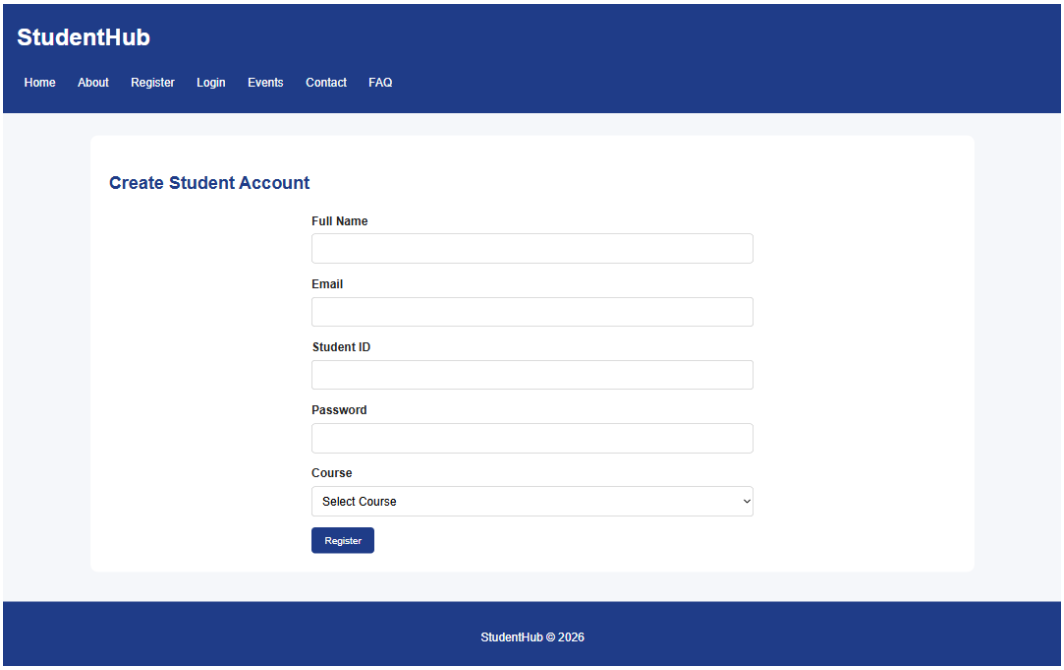
The StudentHub pages were designed using responsive web design principles. CSS Flexbox, CSS Grid, media queries, and Bootstrap provide flexible layouts that adapt to mobile, tablet, and desktop screens. Consistent spacing, typography, colors, and reusable components improve usability and maintainability.

13. Screenshots

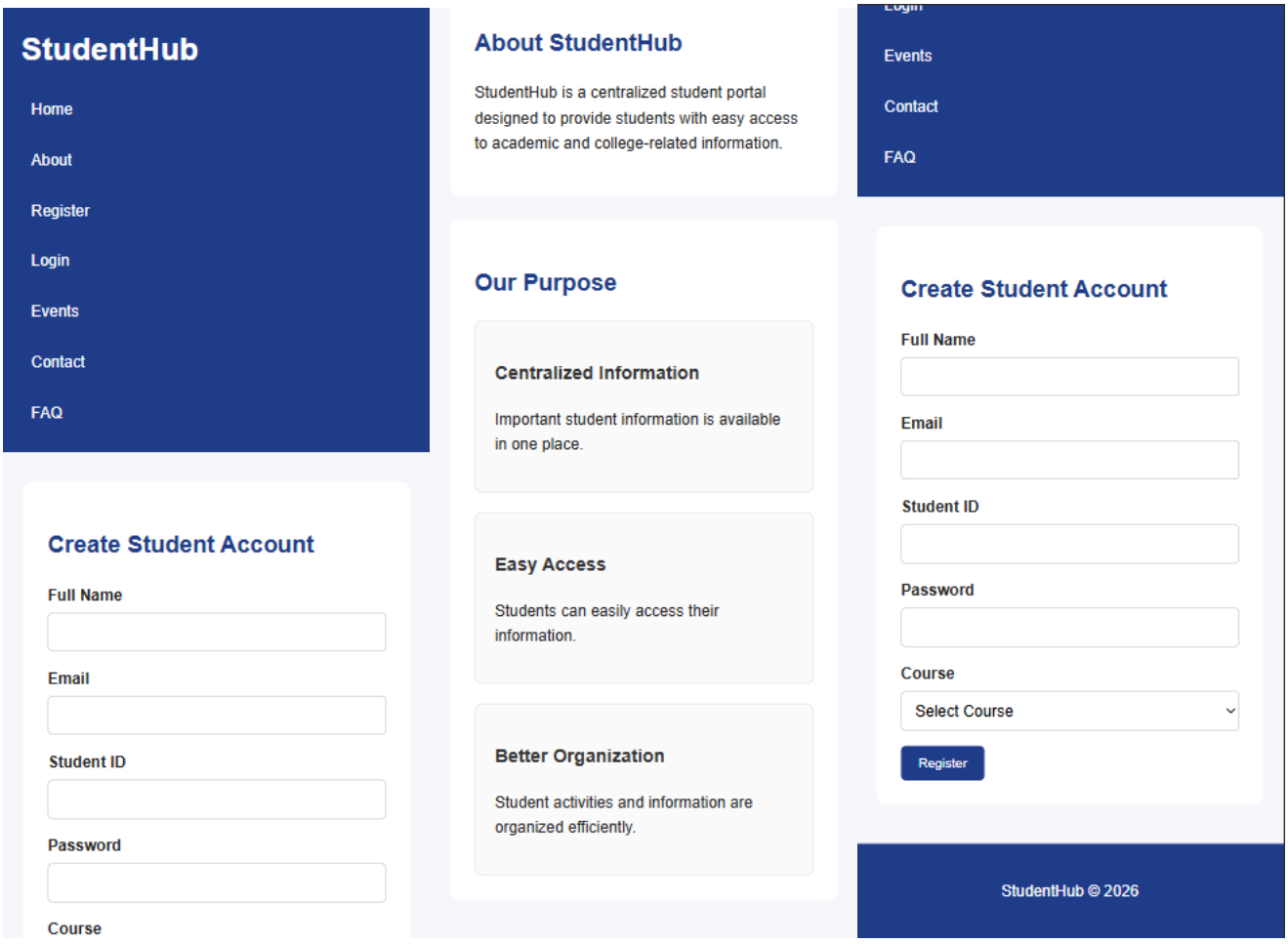
Desktop View:



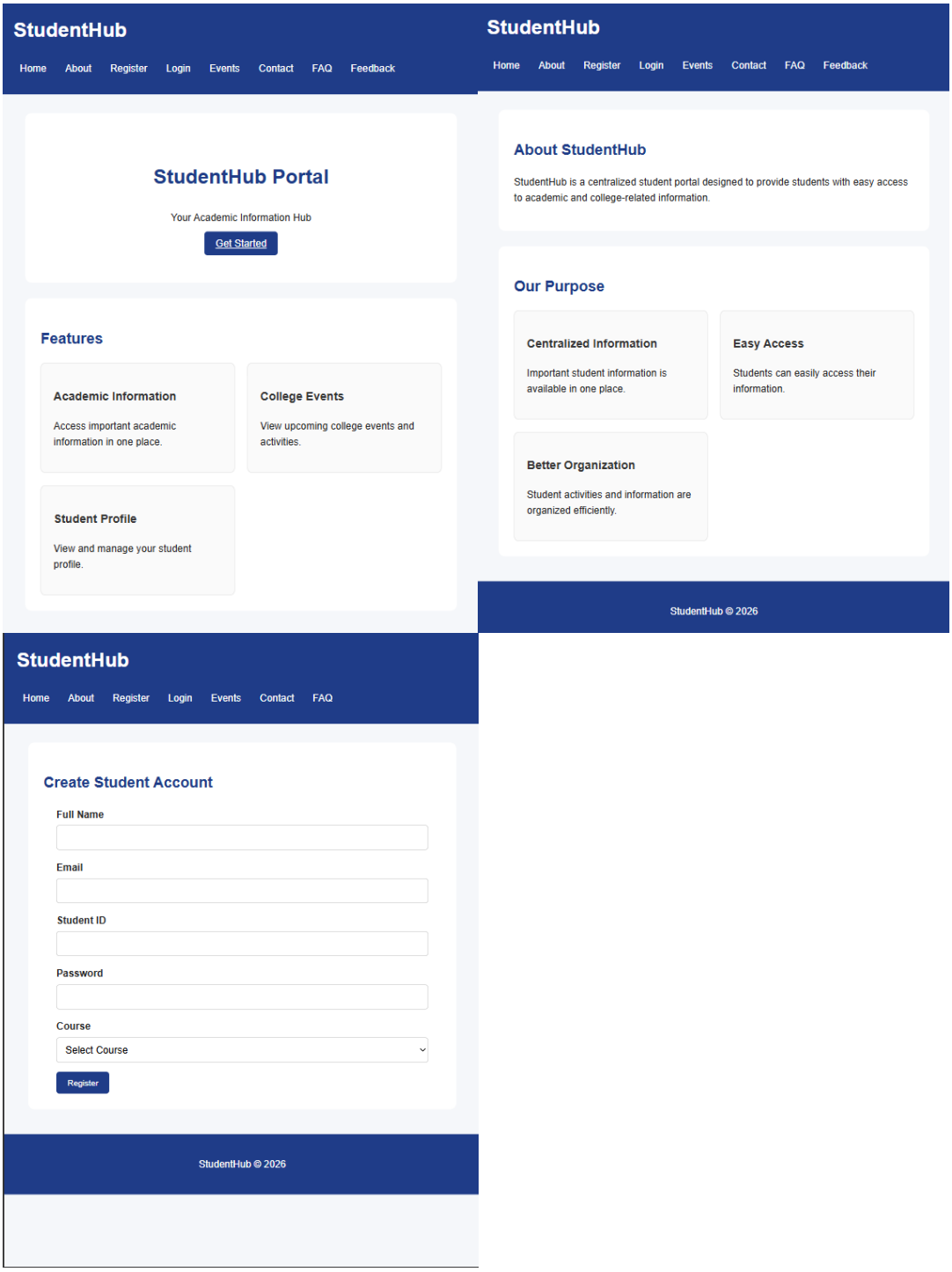



StudentHub
Home
About
Register
Login
Events
Contact
FAQ
Create Student Account
Full Name
Email
Student ID
Password
Course
Select Course
Register
StudentHub © 2026

Mobile View:


StudentHub
Home
About
Register
Login
Events
Contact
FAQ
About StudentHub
StudentHub is a centralized student portal designed to provide students with easy access to academic and college-related information.
Our Purpose
Centralized Information
Important student information is available in one place.
Easy Access
Students can easily access their information.
Better Organization
Student activities and information are organized efficiently.
Events
Contact
FAQ
Create Student Account
Full Name
Email
Student ID
Password
Course
Select Course
Register
StudentHub © 2026

Tablet View:


StudentHub
Home About Register Login Events Contact FAQ Feedback
StudentHub Portal
Your Academic Information Hub
Get Started
Features
Academic Information
Access important academic information in one place.
College Events
View upcoming college events and activities.
Student Profile
View and manage your student profile.
StudentHub
Home About Register Login Events Contact FAQ Feedback
About StudentHub
StudentHub is a centralized student portal designed to provide students with easy access to academic and college-related information.
Our Purpose
Centralized Information
Important student information is available in one place.
Easy Access
Students can easily access their information.
Better Organization
Student activities and information are organized efficiently.
StudentHub © 2026
StudentHub
Home About Register Login Events Contact FAQ
Create Student Account
Full Name
Email
Student ID
Password
Course
Select Course
Register
StudentHub © 2026